Reader: I believe it should be.

Author: You are wrong. Here is a simple illustration:

$$(y_n) = \frac{1}{2}, \frac{2}{3}, \frac{1}{4}, \frac{4}{5}, \frac{1}{6}, \frac{6}{7}, \frac{1}{8}, \ldots\ldots$$

$$(z_n) = \frac{1}{2}, \frac{1}{3}, \frac{3}{4}, \frac{1}{5}, \frac{5}{6}, \frac{1}{7}, \frac{7}{8}, \ldots\ldots$$

→ Brilliant counterexample

$$(y_n + z_n) = 1, 1, 1, 1, 1, 1, 1, \ldots\ldots$$

As you see, the sequences (y_n) and (z_n) are not convergent, while the sequence $(y_n + z_n)$ is convergent, its limit being equal to unity.

Thus, if a sequence $(y_n + z_n)$ is convergent, two alternatives are possible:

sequences (y_n) and (z_n) are convergent as well, or

sequences (y_n) and (z_n) are divergent.

Reader: But can it be that the sequence (y_n) is convergent, while the sequence (z_n) is divergent?

Author: It may be easily shown that this is impossible.

Proof:

Sam: Since, y_n is convergent, we know it must approach a limit a.

By defn, $\forall \varepsilon > 0, \varepsilon \in R, \exists N_1 \in N$ s.t. $\forall n \in N, n > N_1, |y_n - a| < \varepsilon$ (1)

or, $\boxed{a - \varepsilon < y_n < a + \varepsilon}$ (2)

Suppose, for the sake of contradiction, $(y_n + z_n)$ is convergent. Assume it approaches a limit b.

By defn, $\forall \varepsilon > 0, \varepsilon \in R, \exists N_2 \in N$ s.t. $\forall n \in N, n > N_2, |(y_n + z_n) - b| < \varepsilon$ (3)

or, $\boxed{b - \varepsilon < y_n + z_n < b + \varepsilon}$ (4)

Take $N_3 = \max(N_1, N_2)$ $\therefore \forall n \in N, n > N_3$, both (2) and (4) hold.

Subtracting (2) from (4), we get: $\boxed{b - a < z_n < b - a}$ (5)

If we would had taken ε_1 in (2) and ε_2 in (4), we would get:

$$\boxed{(b-a) - (\varepsilon_1 - \varepsilon_2) < z_n < (b-a) + (\varepsilon_1 - \varepsilon_2)} \quad (6)$$

$\therefore |z_n - (b-a)| < (\varepsilon_1 - \varepsilon_2)$ (7). $\therefore z_n$ converges to $(b-a)$ by defn. of limit

This is a contradiction to the fact that z_n is divergent.

$\therefore (y_n + z_n)$ is divergent □